

CTAGS

INDEX YOUR CODE



This article shows how to install and use Ctags to browse through any source code—a very useful tool for Linux programmers.

Ctags, a tool used for code browsing, generates an index file containing entries for each language object in the source code. This index helps search for objects in the source code. A 'tag' is a language object for which there is an entry in the index file.

Ctags supports multiple programming languages—C, C++, C#, COBOL, Flex, Fortran, HTML, Java, JavaScript, Lisp, MATLAB, etc. For a complete list, see <http://ctags.sourceforge.net/languages.html>. It also provides support for user-defined languages with regular expressions.

It generates tags for C/C++ code objects including class names, macro definitions, enumeration names and enumerators, function definitions and prototypes/declarations, classes, interfaces, structs, union data members, structure names, typedefs, union names and variables (definitions and external declarations). It can also print a list of selected objects found in source files. It compiles on UNIX, MSDOS, Windows 95/98/NT, OS/2, QNX, Amiga, QDOS, VMS, Macintosh and Cray.

Installing Ctags

To install, compile from the source. Get the latest version from <http://ctags.sourceforge.net/> and follow the steps below.



Note: The steps to build and configure Ctags on Linux and Cygwin are the same. I built and installed Ctags under Cygwin on Windows.

Go to the Downloads folder and extract the source with `tar -xvf ctags-5.8.tar.gz`. Make a separate directory, `build_ctags`, with `mkdir build_ctags`. Change into the `build_ctags` directory, and then configure the source by running the configure utility in the extracted source. For example: `/cygdrive/c/Users/root/Downloads/ctags-5.8/configure`

To build Ctags, next run 'make', followed by (as the root user) `make install` to install it. Check if the `ctags` command is available (on Windows+Cygwin, look for `ctags.exe` in the folder '`/cygwin/usr/local/bin`'). If yes, the build was successful.

Using Ctags

The use of Ctags is illustrated by browsing through the source code of GCC. First, you need to generate a tag file. Change to the directory containing the source files you want to index. In the case of our example, go to the GCC source code with `cd /cygdrive/d/source_gcc/gcc-4.7.0/gcc/`

Next, generate the tag file with the command `ctags -R *`, which will generate the tag file (named `tags`).

To use the `vi` editor with Ctags, start it up, and then, to find a specific tag, use the `:ta <tag>` command. For example, in Figure 1, the tag '`find_mode`' has been specified. This command should take you to the definition of the tag, as you can see in Figure 2.

To find all occurrences of the particular tag in the index, use the `:ts` command; the result, listing all the occurrences of the tag you searched for with `:ta`, is shown in Figure 3. To go to the location of a result, just type the option number and hit `Enter`.

Tagging your way through source code

Use Ctags to navigate your way through source code:

- If you want to jump to the definition of any other tag used in a particular function definition, use `CTRL+J`.
- `Ctrl+T` is used to step back to the location from where you started your search in the particular directory.

I hope this is enough to start using Ctags for browsing your source code. In the next article, we will discuss another tool, Cscope, which is used for the same purpose, and

(Continued On Page 80...)